Raihan Nur Daffa

Jakarta Barat, Indonesia
085875866779
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https://www.linkedin.com/in/raihan-nur-daffa-1b9552270/

EXPERIENCE

Telkom Indonesia — *Pijar Sekolah (Intern)*
September – December 2025

- Build internal websites for practice and team projects.
- Create website using React js, Tailwind Css
- Participate in fixing bugs/errors on internal websites.
- Conduct basic testing (manual tests or simple integration tests).
- Document issues and solutions for learning purposes.
- Learn how to connect applications with APIs (simulated Telkom APIs or public APIs).

Putra Digital — *Designer (PKL)*
Oktober – November 2021

- Create design for digital marketing
- Know about the industrial world in the field of design
- Make design according to costumer wishes
- Makes Banner – logo – flayers and others

Organization

Himti Mercu Buana — Research and technology division
Oktober 2024

Building the potential of Informatics Engineering students in improving and developing relevant skills, as well as becoming a forum for Mercu Buana Informatics Engineering students to innovate.

Mercu Buana English Club — NewsCast Devision
Oktober 2024

Learning English as a second language and public speaking

Google Developer Student Clubs Mercu Buana University — Member - *February 2024*
I became a member of GDSC so that I could gain many relationships and knowledge about the learning that I am interested in.

LANGUAGES

Indonesia, English (Beginner)

EDUCATION

SMK ISLAM ANNUR LOSARI — *Multimedia*
Januari 2020 – Mei 2023

Universitas Mercu Buana — Teknik Informatika
Septermber 2023 - until now

DEVOLOPER SKILLS

Java Script
Java
Kotlin
Laravel
Python
HTML 5
Cascading Style Sheets (CSS)
PHP
Bootstrap
UI/UX – Figma
Flutter

DATA SCIENCE SKILL

Machine Learning:
Regression, Clustering, Model Evaluation, Feature Engineering, Confosion Matrix, ROC & AUC

NLP: HuggingFace, Sentiment Analysis

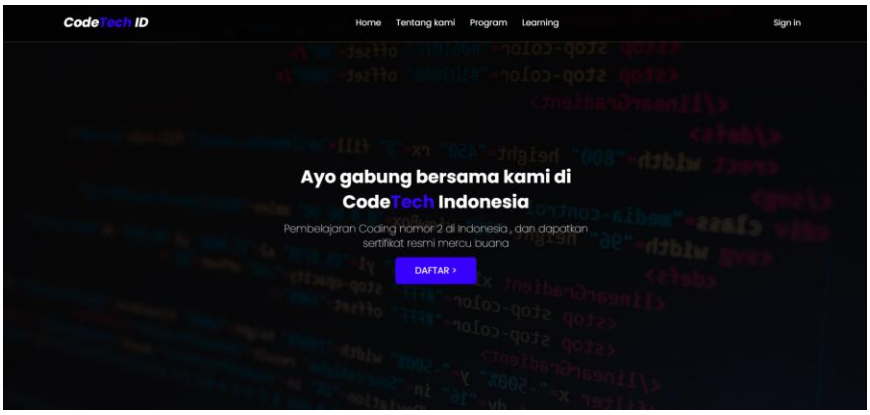
Deep Learning: Framework DL, (CNN), (LSTM), (GANs)

Cloud Computing: Flask, Docker, AWS (basic

PROJECT PORTOFOLIO

CodeTech Indonesia – Website Belajar Online (This website is still in the development stage)

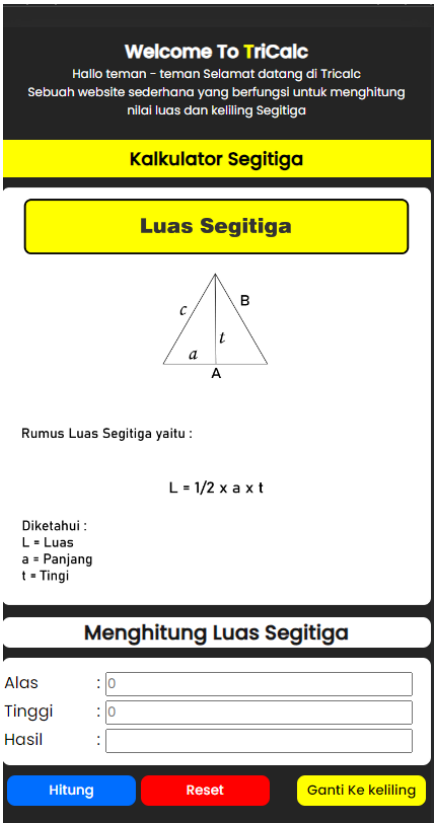
Code ID is a website that aligns education with developments in science and technology to hone practical skills that are more relevant to the world of work. with learning paths will help you learn and hone your skills to become a computer expert



PHP · HTML5 · Cascading Style Sheets (CSS) · JavaScript · Riect Front - End Web · User Interface Design · Front-End Development

Tricalc – Calculator https://revou-fundamental-course.github.io/25-dec-23-RndHanz/

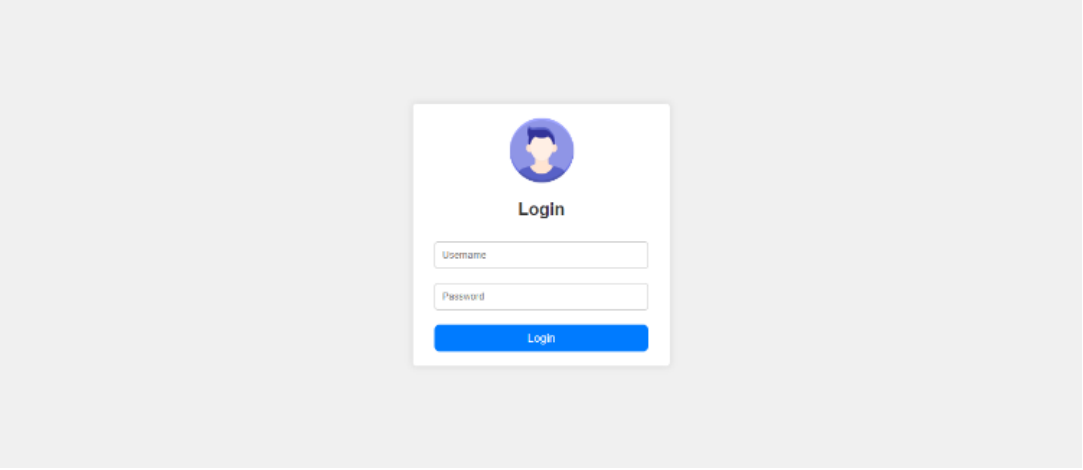
Tricalc A simple website that serves to calculate value of area and circumference of the Triangle - I created this website to complete a mini project assignment from the fundamental class of the RevoU Software Engineer course

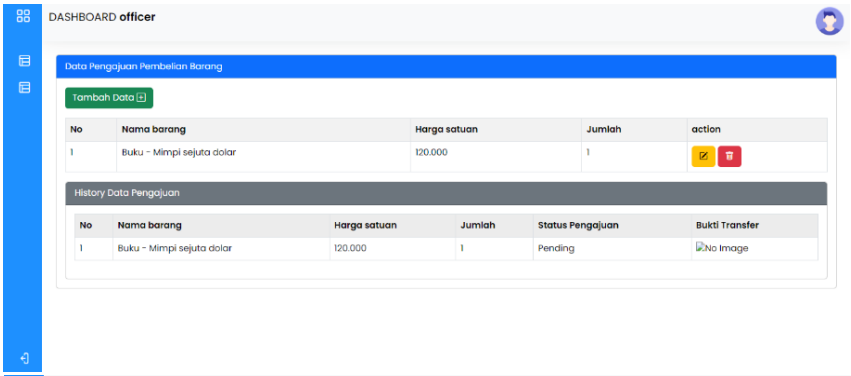


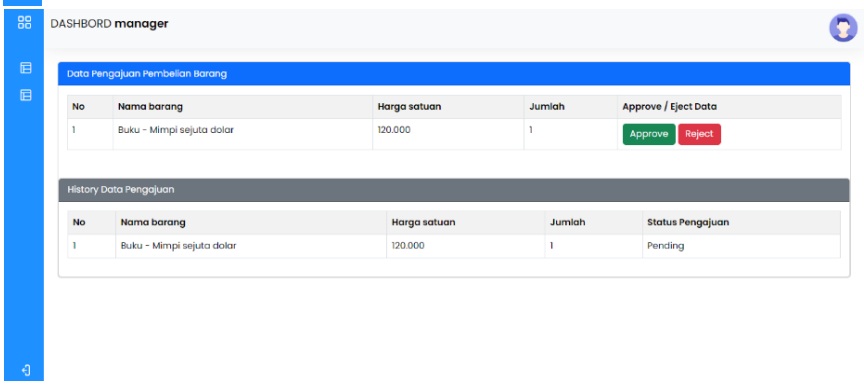
GitHub · HTML5 · Cascading Style Sheets (CSS) · JavaScript

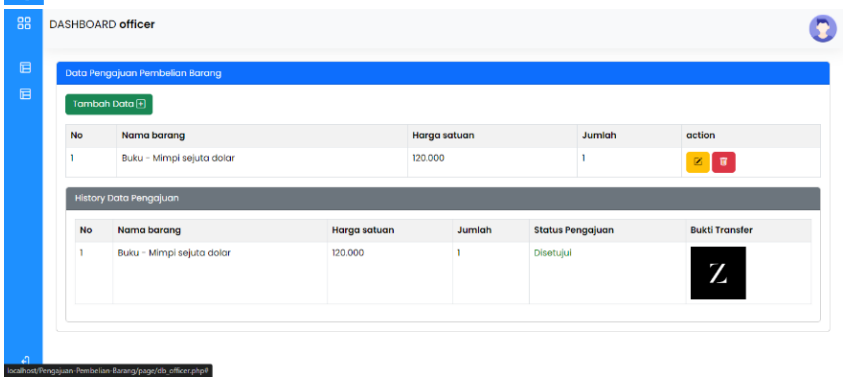
Database – Pengajuan Pembelian Barang (Localhost)
https://github.com/RndHanz/Web-Application-Pengajuan-Pembelian-Barang

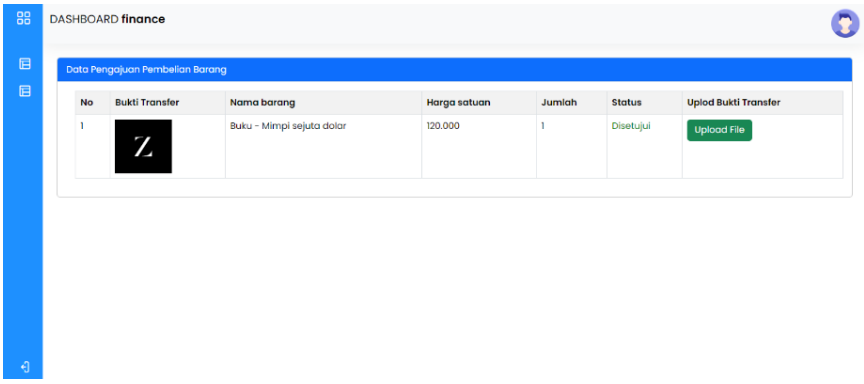
This website is for 3 databases, including officer, manager, finance, where each database has its own function, officer to add products then the manager approves the product and finance adds proof of payment if approved by the manager







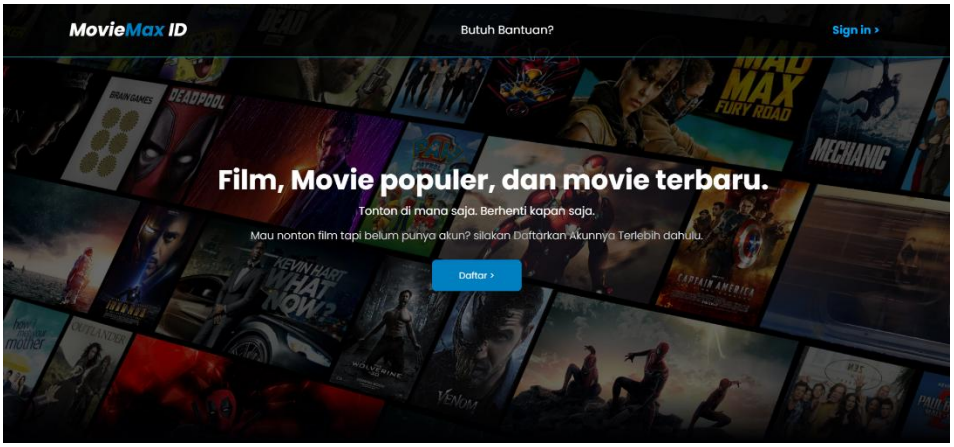


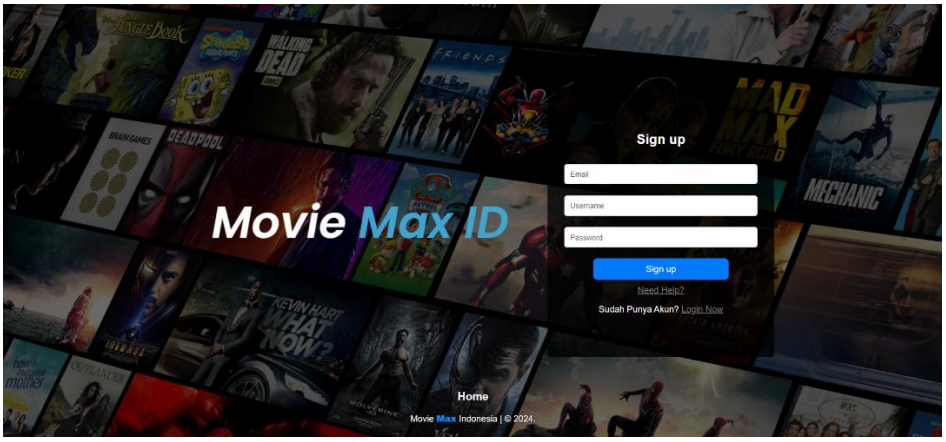


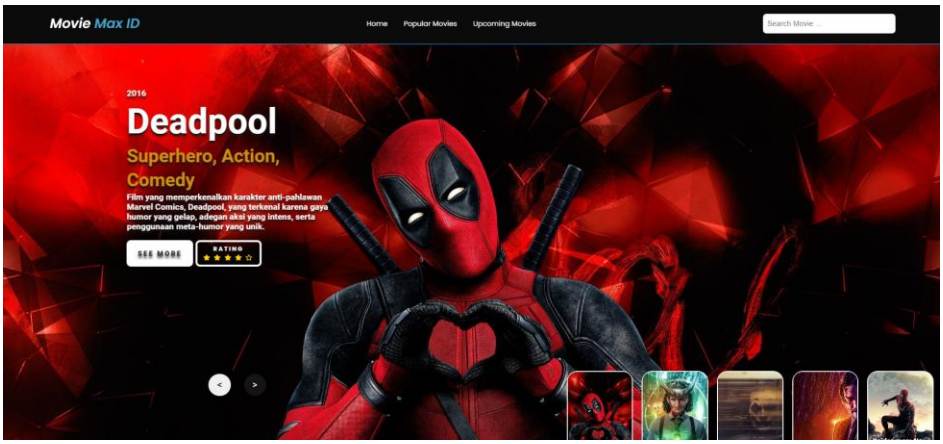
MySQL · PHP · HTML5 · Cascading Style Sheets (CSS) - Bootstrap

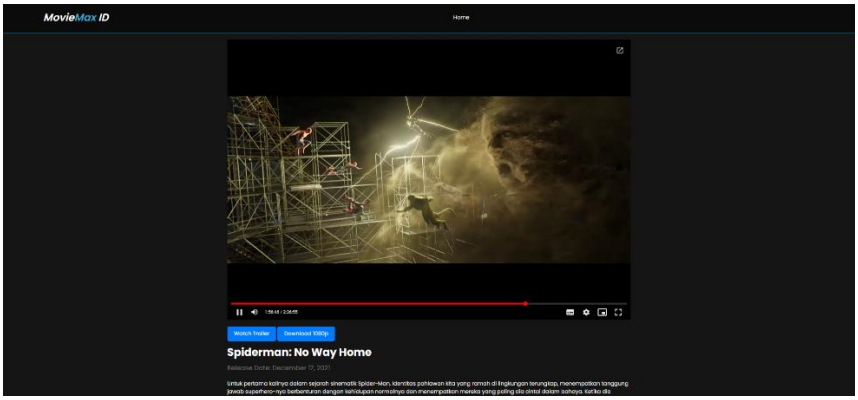
Movie Max ID – (Localhost) Not Public

This website was created to make it easier to watch films anywhere, anytime and for free. Of course, we created this for pirated viewers.





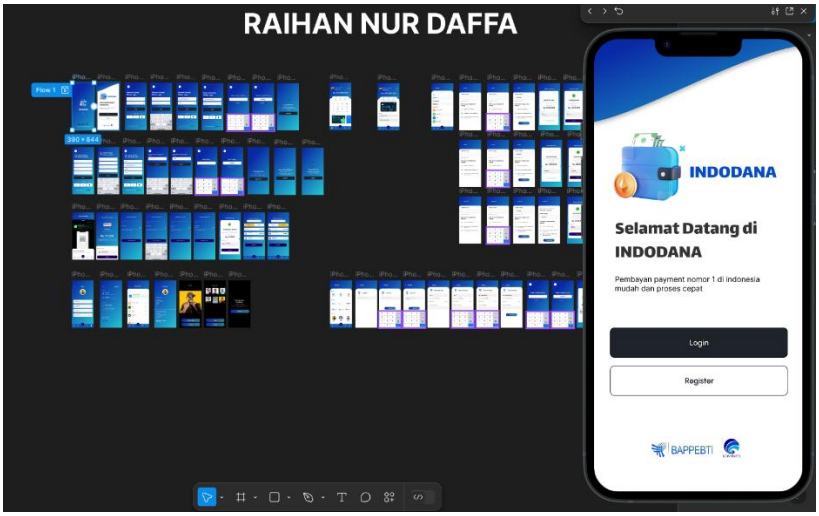


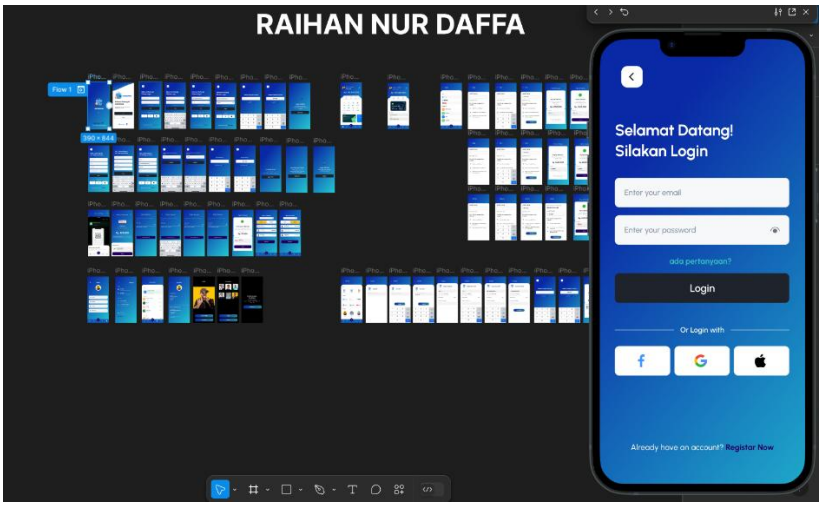


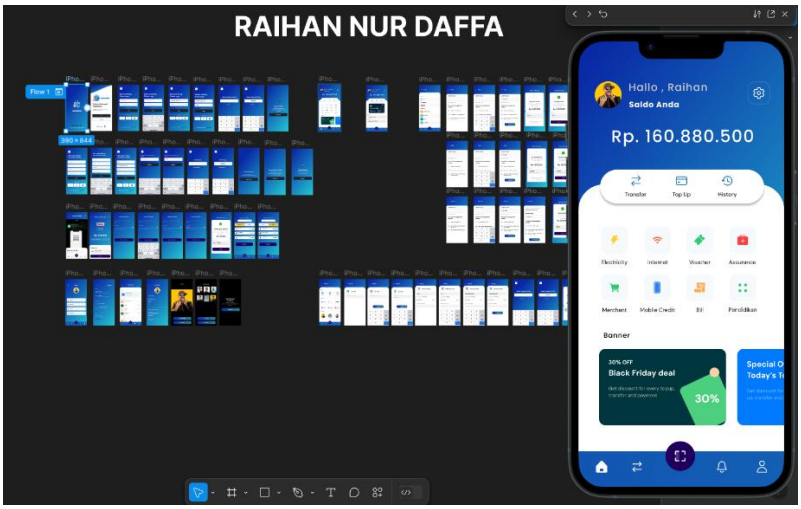
Database · PHP · HTML5 · Cascading Style Sheets (CSS) - Bootstrap – use API Movie

E-Wallet Mobile Apps – Figma

I created this mobile app UI as a final project for UI/UX bootcamp at Himti Uin https://www.figma.com/design/LfCqgN0ESIJuVkd1wmgIol/E-Wallet-Mobile-Apps---INDODANA--Community-?node-id=0-1&t=9GKhbIeFRXHnaLZu-1

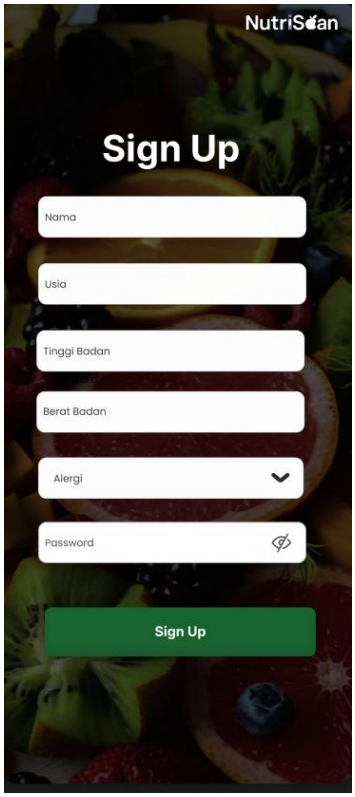


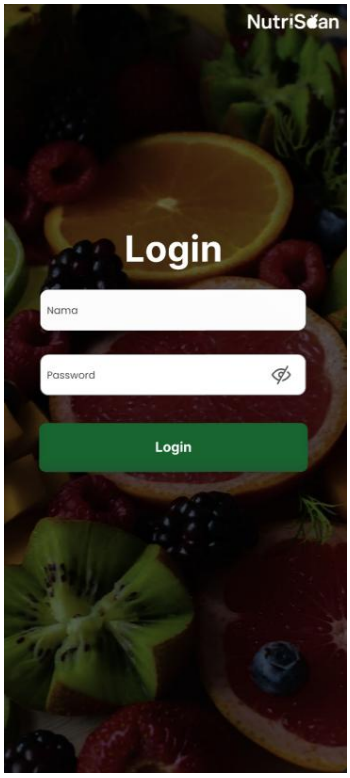


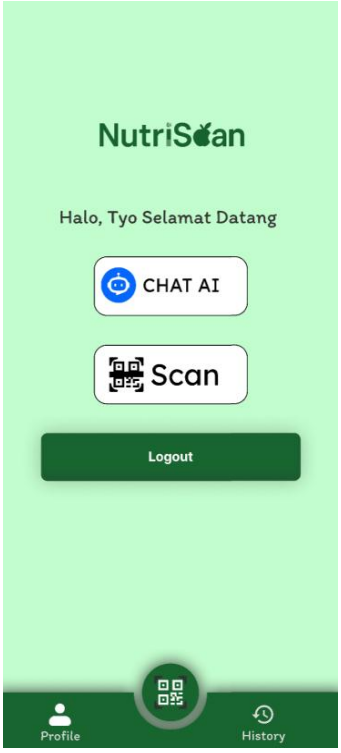


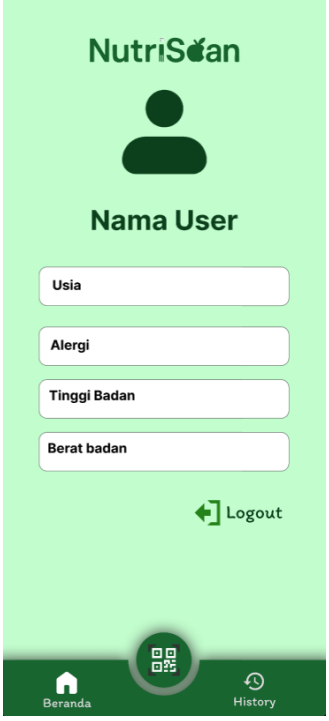
NutriScan – Android Devoloper
https://github.com/RndHanz/NutriScanAPP

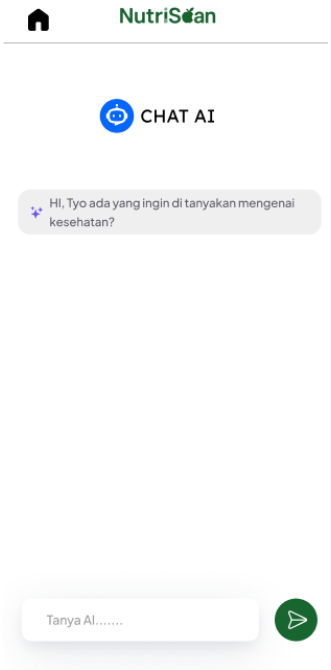
I created a Nutriscan app project that can detect food allergies using the scanning feature and I created it in Android Studio.







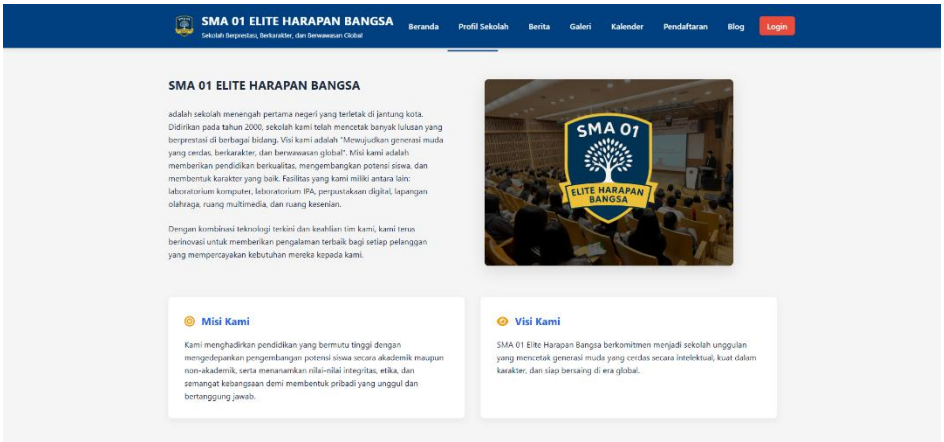


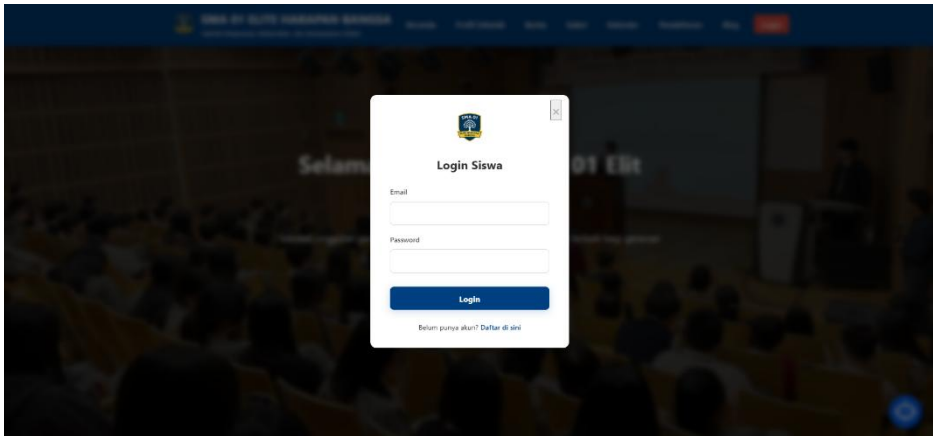


Database · PHP · front end use Java and use API

Sistem Manajemen Sekolah – **Web Devoloper**
https://github.com/RndHanz/Sistem-Manajemen-sekolah

I created a Nutriscan app project that can detect food allergies using the scanning feature and I created it in Android Studio.

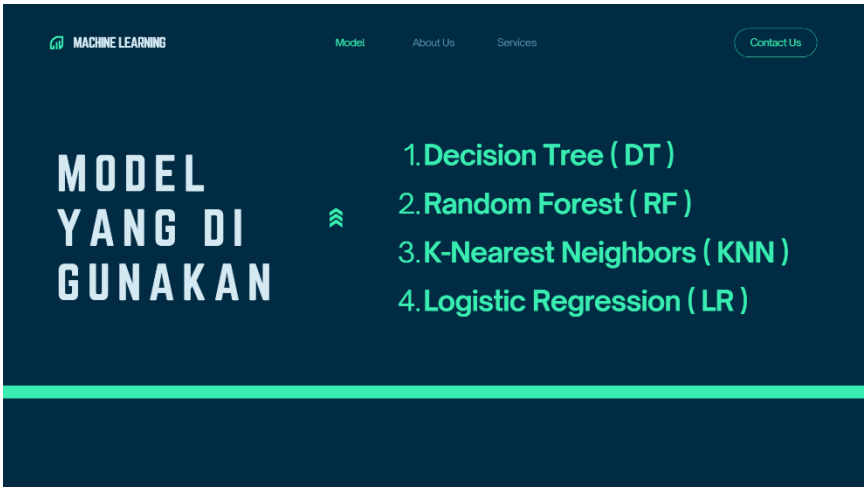


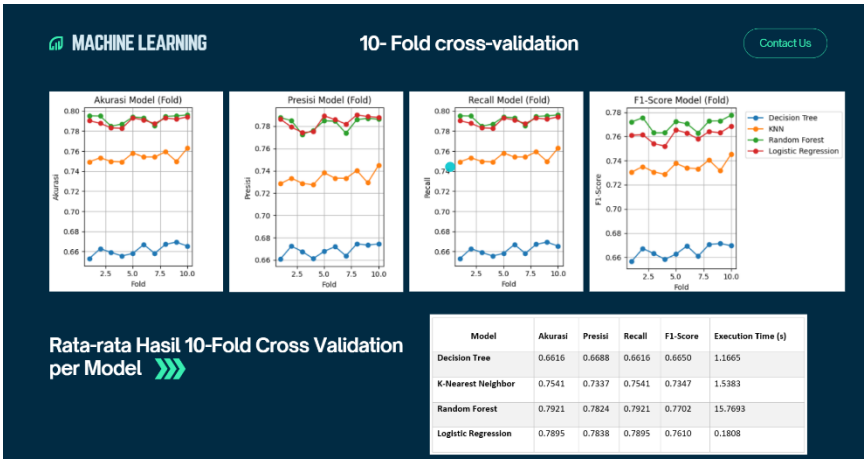


Database · PHP · front end Html,css,javascript and use API

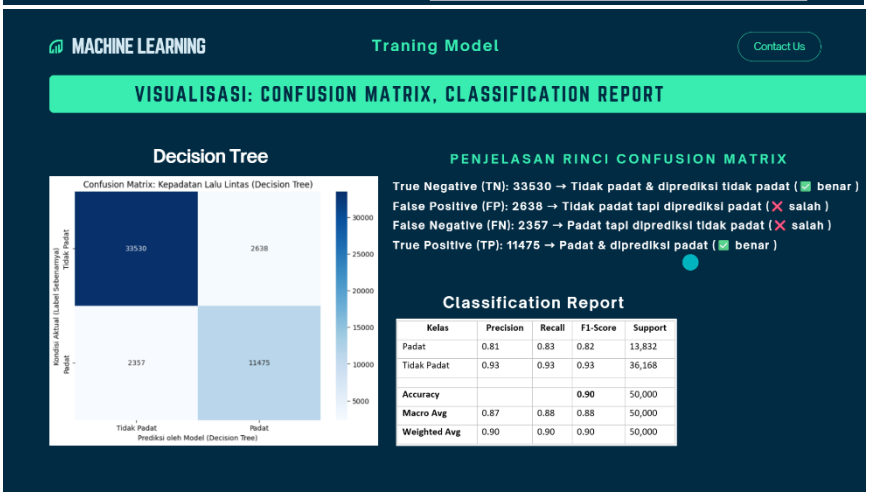
Predictive Analytics for Traffic Density: A Machine Learning Approach – Data Science (Machine Learning)

This is traffic density using machine learning model, and using some libraries from python, and we use artificial dataset from gpt chat.

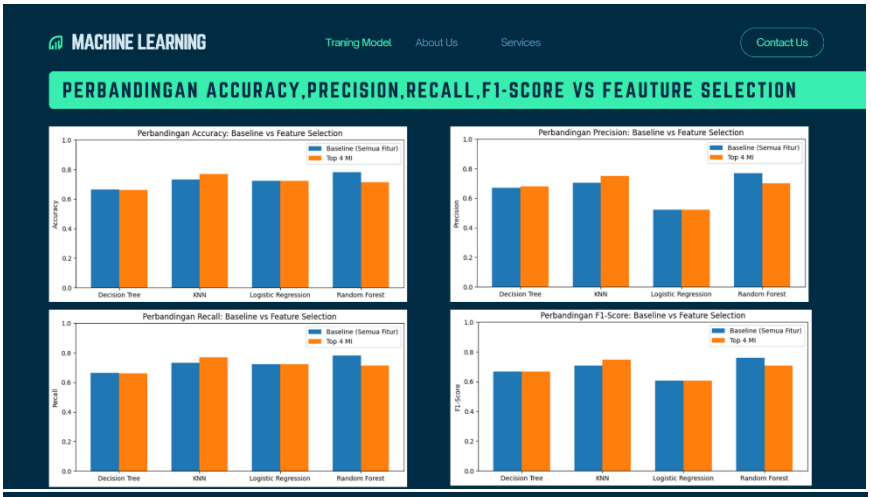


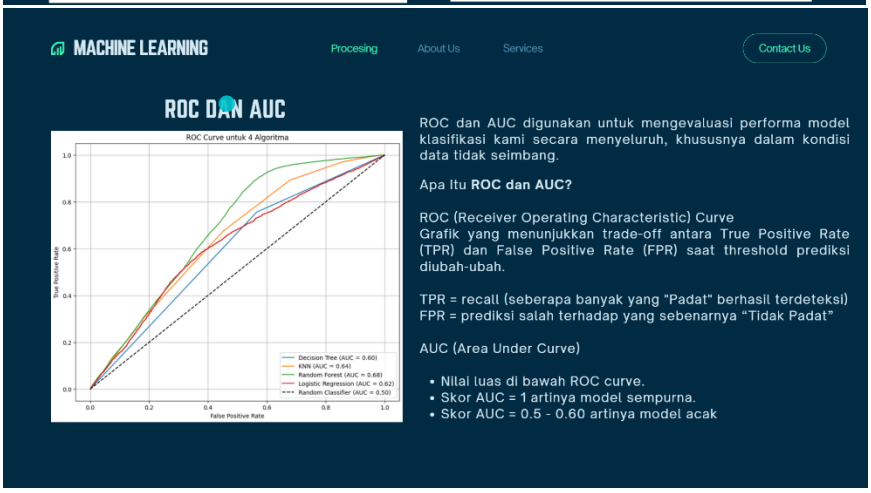


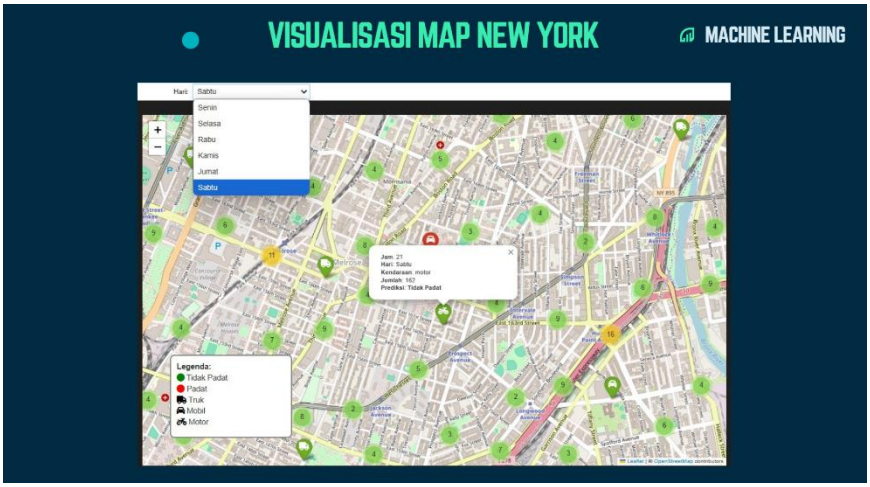
Model	Akurasi	Presisi	Recall	F1-Score	Execution Time (s)
Decision Tree	0.6616	0.6688	0.6616	0.6650	1.1665
K-Nearest Neighbor	0.7541	0.7337	0.7541	0.7347	1.5383
Random Forest	0.7921	0.7824	0.7921	0.7702	15.7693
Logistic Regression	0.7895	0.7838	0.7895	0.7610	0.1808



Kelas	Precision	Recall	F1-Score	Support
Padat	0.81	0.83	0.82	13,832
Tidak Padat	0.93	0.93	0.93	36,168
Accuracy			**0.90**	50,000
Macro Avg	0.87	0.88	0.88	50,000
Weighted Avg	0.90	0.90	0.90	50,000







Library Python:

Pandas, NumPy, Matplotlib.

Machine Learning:

Regression, Clustering, Model Evaluation, Feature Engineering,10-cross Validation, Confusion Matrix

Analisis Efesiensi Komputasi dan skaliblitas algoritma lightgbm pada kelasifikasi varietes kacang kering – Data Science (Machine Learning, computer vision)

This is machine learning model and computer vision, using some libraries from python

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DOI: 10.25126/jtiik...
p-ISSN: 2355-7699
e-ISSN: 2528-6579

ANALISIS EFISIENSI KOMPUTASI DAN SKALABILITAS ALGORITMA LIGHTGBM PADA KLASIFIKASI VARIETAS KACANG KERING

Raihan Nur Daffa[*1], Muhamad Febrian Putra[2], Zaidan Arasyid Suwondo[3]

[1,2,3]Universitas Mercu Buana, Jakarta Barat Meruya
Email: [1]41523010203@student.mercubuana.ac.id, [2]41523010210@student.mercubuana.ac.id,
[3]41523010299@student.mercubuana.ac.id
[*]Penulis Korespondensi

(Naskah masuk: dd mmm yyyy, diterima untuk diterbitkan: dd mmm yyyy)

Abstrak

Klasifikasi varietas kacang kering merupakan proses penting dalam bidang pertanian dan pengendalian kualitas pangan. Namun, metode identifikasi secara manual masih memiliki keterbatasan dari sisi efisiensi dan konsistensi akibat kemiripan karakteristik antar varietas kacang. Selain tingkat akurasi, efisiensi komputasi dan skalabilitas model menjadi aspek penting dalam pemilihan algoritma klasifikasi, terutama untuk penerapan pada sistem berbasis aplikasi dengan sumber daya terbatas. Penelitian ini bertujuan menganalisis efisiensi komputasi dan skalabilitas algoritma LightGBM pada klasifikasi varietas kacang kering. Dataset Dry Bean digunakan sebagai data utama, dengan tahapan pra-pemrosesan meliputi pemeriksaan kualitas data dan normalisasi fitur. Evaluasi performa model dilakukan menggunakan metrik *accuracy*, *precision*, *recall*, dan *F1-score*, sedangkan efisiensi komputasi dianalisis berdasarkan waktu pemrosesan model. Aspek skalabilitas dianalisis secara deskriptif dengan mengamati kestabilan kinerja model terhadap karakteristik data yang digunakan. Hasil pengujian menunjukkan bahwa algoritma LightGBM mampu menghasilkan performa klasifikasi yang baik dengan waktu komputasi yang efisien dan stabil. Karakteristik pembelajaran berbasis histogram pada LightGBM mendukung pemrosesan data secara cepat sehingga sesuai untuk dataset berukuran besar. Hasil penelitian ini menunjukkan bahwa LightGBM merupakan algoritma yang efektif untuk klasifikasi varietas kacang kering.

Kata kunci: *klasifikasi kacang kering, LightGBM, efisiensi komputasi, skalabilitas, machine learning.*

ANALYSIS OF COMPUTATIONAL EFFICIENCY AND SCALABILITY OF THE LIGHTGBM ALGORITHM FOR DRY BEAN VARIETY CLASSIFICATION

Abstract

Dry bean variety classification is an important task in agriculture and food quality control. However, manual identification methods still suffer from efficiency and consistency limitations due to similarities among bean varieties. In addition to classification accuracy, computational efficiency and scalability are critical factors when selecting classification algorithms, particularly for application-based systems with limited computational resources. This study aims to analysis the computational efficiency and scalability of the Light GBM algorithm for dry bean variety classification. The Dry Bean dataset is used as the primary data source, with preprocessing steps including data quality checking and feature normalization. Model performance is evaluated using accuracy, precision, recall, and F1-score metrics, while computational efficiency is assessed based on model processing time. Scalability is analysis descriptively by observing the stability of model performance with respect to the characteristics of the dataset used. Experimental results indicate that Light GBM achieves good classification performance with efficient and stable computational time. The histogram-based learning mechanism employed by Light GBM enables fast data processing, making it suitable for large-scale datasets. as a means of implementation and result visualization. The results of this study demonstrate that Light GBM is an effective and scalable algorithm for dry bean variety classification.

Keywords: dry bean classification, Light GBM, computational efficiency, scalability, machine learning.

UNIVERSITAS MERCU BUANA

ANALYSIS OF COMPUTATIONAL EFFICIENCY AND SCALABILITY OF THE LIGHTGBM ALGORITHM FOR DRY BEAN VARIETY CLASSIFICATION

Raihan Nur Daffa[*1], Muhamad Febrian Putra[2], Zaidan Arasyid Suwondo[3]
[1,2,3]Universitas Mercu Buana, Jakarta Barat Meruya
[1]41523010203@student.mercubuana.ac.id

Introduction

Dry bean variety classification plays an important role in agriculture and food quality control. Manual identification is often inefficient and inconsistent due to the high similarity of morphological characteristics among bean varieties. Machine learning approaches provide an effective alternative by enabling automatic and objective classification. In addition to prediction accuracy, computational efficiency and scalability are critical factors, especially for real-world applications with limited computational resources. This study focuses on evaluating the computational efficiency and scalability of the LightGBM algorithm for dry bean variety classification.

Dataset

The experiment uses the Dry Bean Dataset, which contains 13,611 samples, 16 numerical features, and 7 dry bean varieties. The features represent morphological characteristics such as area, perimeter, axis lengths, and shape factors, making the dataset suitable for multiclass classification using machine learning techniques.

Results

The experimental results indicate that LightGBM achieves strong classification performance across all dry bean varieties. The overall accuracy exceeds 90%, with balanced precision, recall, and F1-score values for each class. The confusion matrix shows low misclassification rates, confirming the model's ability to effectively distinguish between bean varieties. Additionally, the stable performance across different data sizes highlights the robustness of the model.

Concolusion

LightGBM achieves high and stable accuracy with efficient processing time for dry bean variety classification. Its scalability and computational efficiency make it a reliable solution for numerical-based agricultural classification tasks.

References

Fang, W., Zhang, Y. & Liu, H. (2021). Computational efficiency of LightGBM and XGBoost. Journal of Big Data.
Krishnan, S. et al (2023). Dry bean variety identification using machine learning. Scientific Programming.
Liu, T., Wang, Y. & Zhou, J. (2023). Scalability of tree-based ensemble models. Information Sciences.

Methods

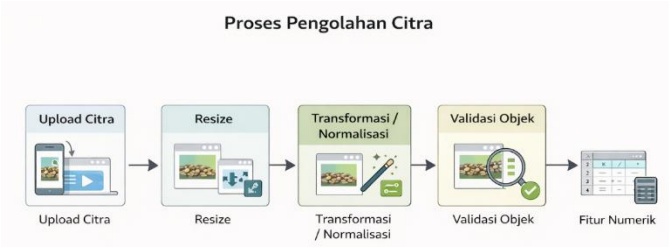
The research methodology follows an experimental approach consisting of data preprocessing, model training, and performance evaluation. The dataset is first examined to ensure data quality, followed by outlier handling and feature normalization. The processed data is then divided into training and testing sets. A LightGBM model is trained using the training data and evaluated on unseen test data. Performance is assessed using classification metrics, while computational efficiency and scalability are analyzed based on processing time and model stability across different data sizes.

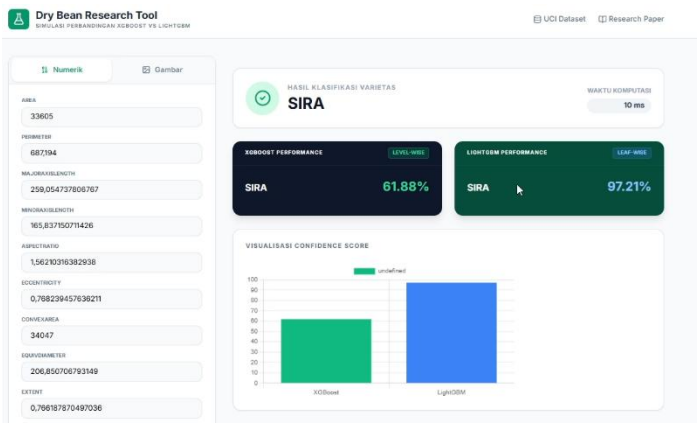
Computational Efficiency

Computational efficiency is evaluated based on training and testing time. The results indicate that LightGBM achieves fast and stable processing during both training and inference. Its histogram-based learning mechanism reduces computational overhead, making the model suitable for applications requiring efficient and repeated predictions.

Scalability Analysis

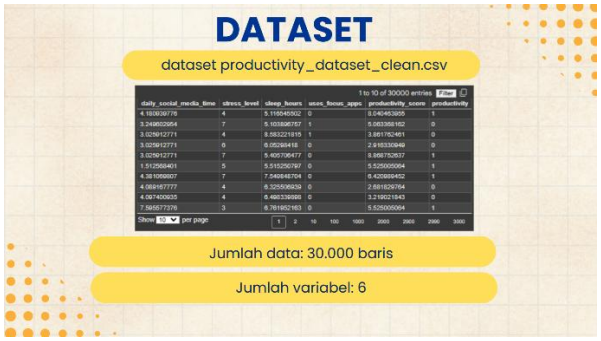
Scalability is evaluated using different dataset sizes (20%–100%). The results show a gradual increase in training time without extreme spikes, while model accuracy improves and remains stable as data size increases. This indicates that LightGBM maintains consistent performance when handling larger datasets.

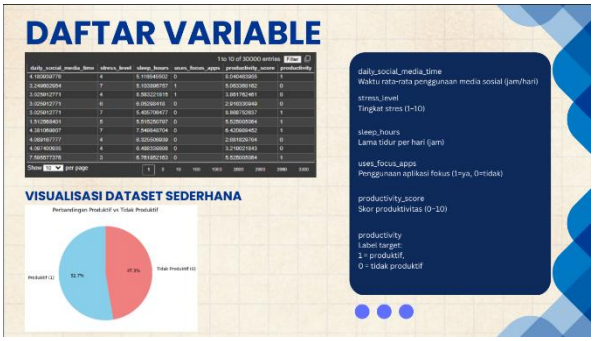


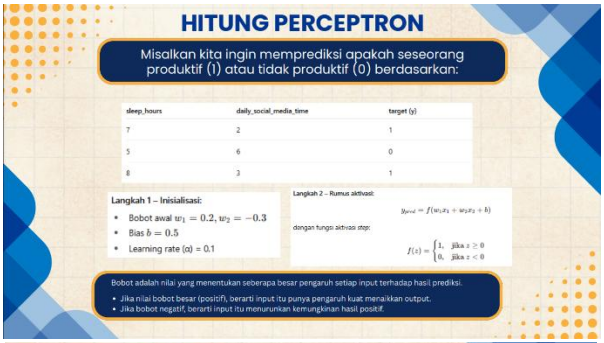


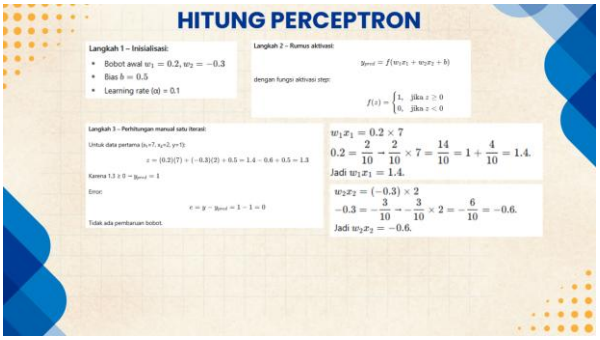
Prediksi Produktivitas berdasarkan kebiasaan harian menggunakan model deep learning – Data Science (Deep Learning)

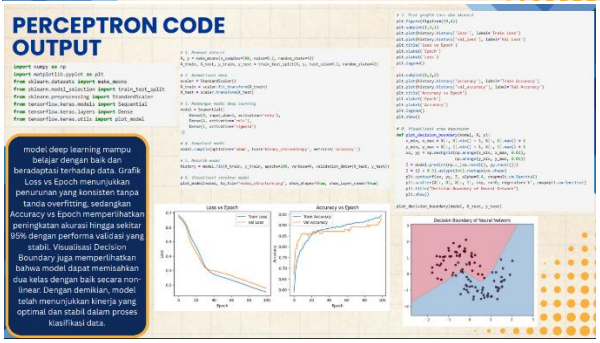
This is Deep Learning model and using some libraries from python











Certificate

SMK Telkom Jakarta & Himti *– Pengabdian Masyarakat menjadi pengajar dengan materi membuat website pertama*

Harisenin.com *– UI/UX Design*

NiagaHoster *– Wordpress*

RevoU *– Fundamental course Sofware Engineer*

Dicoding – Belajar Membuat front end web untuk pemula

Dicoding – Belajar Dasar Pemograman Web

Dicoding – Belajar Dasar Ai (Artificial Intelligence)

HackeRank – Python (Basic)

Dicoding – Belajar Dasar Structured Query Langues (SQL)

Dicoding – Belajar Dasar Visualisasi Data

Dicoding – Cloud Proctitioner Essentials (Belajar Dasar AWS Cloud)

Bootcamp UI/UX Design – HIMTI UIN Syarif Hidayatullah Jakarta

Full Stack Devoloper – HIMTI Mercu Buana

Dicoding – Memulai Pemograman dengan Java

Dicoding – Memulai Pemograman dengan JavaScript